

A Structural Selection Diagnostic on the Riemann Zeta Critical Line

A purely diagnostic, non-proof toy experiment with controls, ablations, and pair-correlation tests

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Abstract

This note applies a simple heuristic structural diagnostic to the nontrivial zeros of the Riemann zeta function. The exercise is explicitly not an attempt to prove or support the Riemann Hypothesis. Instead, it treats the critical line as a well-known mathematical test case for diagnostic scoring rules. The question is whether known zeta zeros appear as low-cost points under the score, and how much of that behaviour is built into the score itself.

The proposed score combines three terms: a zero term $H = (1 + |\zeta(\sigma + it)|)^{-1}$, a critical-line balance term $B = (1 + \alpha|\sigma - 1/2|)^{-1}$, and a local symmetric σ -response term R . The resulting cost is

$$F_{\text{diag}} = -\log(\epsilon + H) - \log(\epsilon + B) - \log(\epsilon + R).$$

The experiment evaluates 40 known zeros, jittered-zero controls, random critical-line controls, ablations without B or without H , and parameter sweeps over α and the robustness displacement. The second main test is a pair-correlation diagnostic: the normalised nearest-neighbour spacings of the first 200 zeta zeros are compared with a GUE nearest-neighbour spacing proxy and with Poisson and uniform controls.

Known zeros exhibit a stable minimum at $\delta = 0$ under the full diagnostic and under the no-balance ablation. However, random critical-line points also inherit a $\delta = 0$ preference when the balance and robustness terms are present. This is the central limitation: known zeros have low diagnostic stress under the proposed score, but part of the critical-line preference is built into the score itself.

The result is a diagnostic exercise, not evidence for the Riemann Hypothesis. Its methodological contribution is to show how a visually persuasive score can be stress-tested against controls, ablations, parameter sweeps, and pair-geometry diagnostics.

1 Purpose and Scope

No Riemann-Hypothesis claim. This paper contains no claim regarding the truth of the Riemann Hypothesis. The observed critical-line minima are partly produced by the construction of the diagnostic itself. The purpose is to test a scoring rule and its controls, not to infer a theorem about the zeta function.

The Riemann Hypothesis states that all nontrivial zeros of the zeta function lie on the critical line

$$\text{Re}(s) = \frac{1}{2}. \tag{1}$$

The present note does not attempt to prove or support this statement in the mathematical sense. Instead, it asks a weaker diagnostic question:

Do known zeta-zero locations appear as minima of a simple diagnostic cost when compared with small off-line displacements?

This is useful only if controls are included. Without controls, the test is nearly tautological: a score that rewards $|\zeta(s)| \simeq 0$ and explicitly rewards $\sigma = 1/2$ will naturally favour known zeros on the critical line. For this reason, the experiment includes ablations and random critical-line points.

2 Diagnostic Score

Let

$$s = \sigma + it. \quad (2)$$

The first term rewards closeness to a zeta zero:

$$H(\sigma, t) = \frac{1}{1 + |\zeta(\sigma + it)|}. \quad (3)$$

This term is intentionally direct. It is also the most dangerous term from the standpoint of overinterpretation, because known zeros are selected precisely by small $|\zeta|$.

The second term encodes balance around the critical line:

$$B(\sigma) = \frac{1}{1 + \alpha|\sigma - \frac{1}{2}|}. \quad (4)$$

This term explicitly favours $\sigma = 1/2$. It is therefore not independent evidence for the critical line.

The third term is a local symmetric-response term:

$$R(\sigma, t) = \frac{1}{1 + |\zeta(\sigma + \Delta\sigma + it) - \zeta(\sigma - \Delta\sigma + it)|}. \quad (5)$$

It measures local sensitivity under small real-direction displacement.

The full diagnostic cost is

$$\boxed{F_{\text{diag}}(\sigma, t) = -\log(\epsilon + H) - \log(\epsilon + B) - \log(\epsilon + R)}. \quad (6)$$

Lower cost means lower diagnostic stress under the proposed score.

3 Controls and Ablations

The benchmark uses three point sets:

- known zeros $1/2 + it_n$ computed with `mpmath.zetazero`,
- jittered zeros $t_n + \eta_n$ with small Gaussian noise,
- random critical-line points sampled across the same t range.

For each point, the score is evaluated at

$$\delta \in \{-0.05, -0.02, -0.01, -0.005, -0.002, 0, 0.002, 0.005, 0.01, 0.02, 0.05\}, \quad (7)$$

where $\sigma = 1/2 + \delta$.

Three score configurations are compared:

Configuration	Meaning
HBR	full diagnostic
HR	no explicit balance-to-critical-line term
BR	no direct zero term

The default parameters are

$$\alpha = 10, \quad \Delta\sigma = 0.002, \quad \epsilon = 10^{-12}. \quad (8)$$

The sweep varies

$$\alpha \in \{0, 2, 5, 10, 20\}, \quad \Delta\sigma \in \{0.001, 0.002, 0.005\}. \quad (9)$$

4 Results

4.1 Default Minima

The default run uses 40 known zeros and 40 random critical-line controls.

Table 1: Best mean off-line shift under the default diagnostic.

Point set	Config	Best δ	Mean F at best
Known zeros	HBR	0.000	0.00818
Known zeros	HR	0.000	0.00818
Known zeros	BR	0.000	0.00818
Jittered zeros	HBR	0.000	0.16544
Jittered zeros	HR	0.005	0.16509
Jittered zeros	BR	0.000	0.00817
Random critical-line	HBR	0.000	0.86054
Random critical-line	HR	0.050	0.83426
Random critical-line	BR	0.000	0.00935

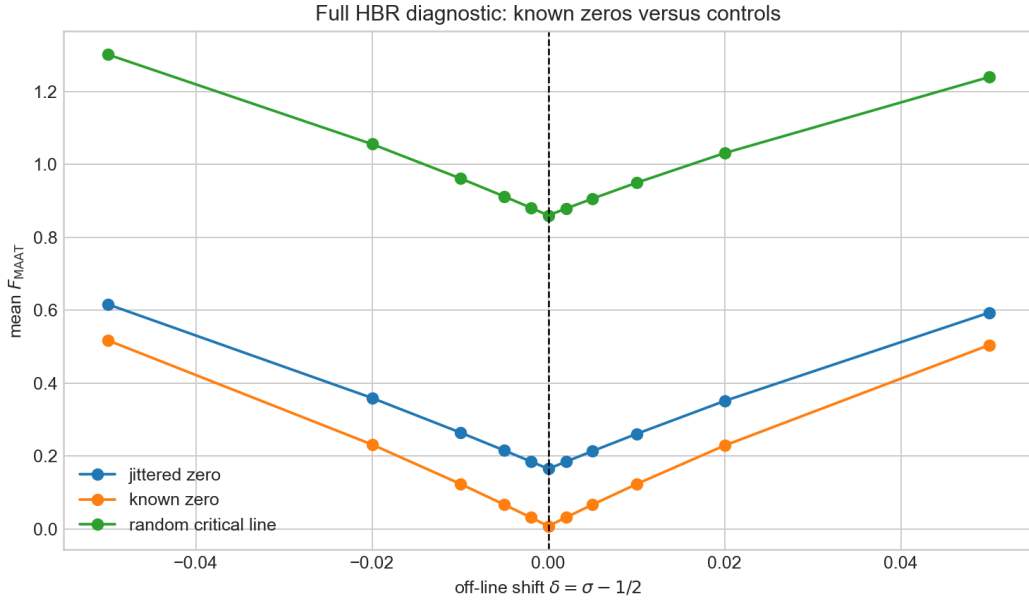


Figure 1: Full HBR diagnostic for known zeros, jittered zeros, and random critical-line points. Known zeros have very low cost at $\delta = 0$, but random critical-line points also inherit a balance-driven $\delta = 0$ minimum.

The full HBR diagnostic strongly favours known zeros on the critical line. The crucial caveat is that the random critical-line controls are also minimised at $\delta = 0$ when the balance term is included. Thus the diagnostic cannot be interpreted as independent evidence for the critical line.

4.2 Ablations

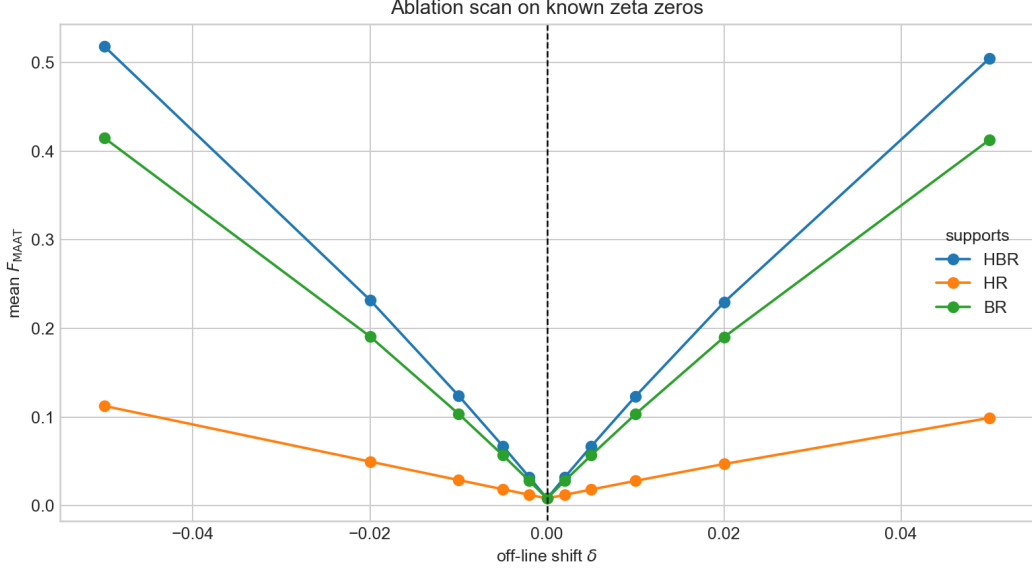


Figure 2: Ablation scan for known zeros. The no-balance HR diagnostic still has its mean minimum at $\delta = 0$, while the no-H BR diagnostic mostly reflects balance and local robustness.

The no-balance HR result is the most informative part of the benchmark. For known zeros, $\delta = 0$ remains the best mean displacement even without the explicit B term. This indicates that the zero and local response terms alone favour the critical-line location of the known zeros.

The no-H BR result is weaker and should be interpreted especially cautiously, because BR largely probes symmetry around the imposed critical-line anchor. It still often favours $\delta = 0$, but this is expected because the balance term remains present and the robustness term is symmetric around the real direction.

4.3 Parameter-Sweep Stability

Table 2: Parameter-sweep stability for known zeros.

Config	Fraction with $\delta = 0$ best	Mean $ \delta_{\text{best}} $
HBR	1.0	0.000
HR	1.0	0.000
BR	0.8	0.010

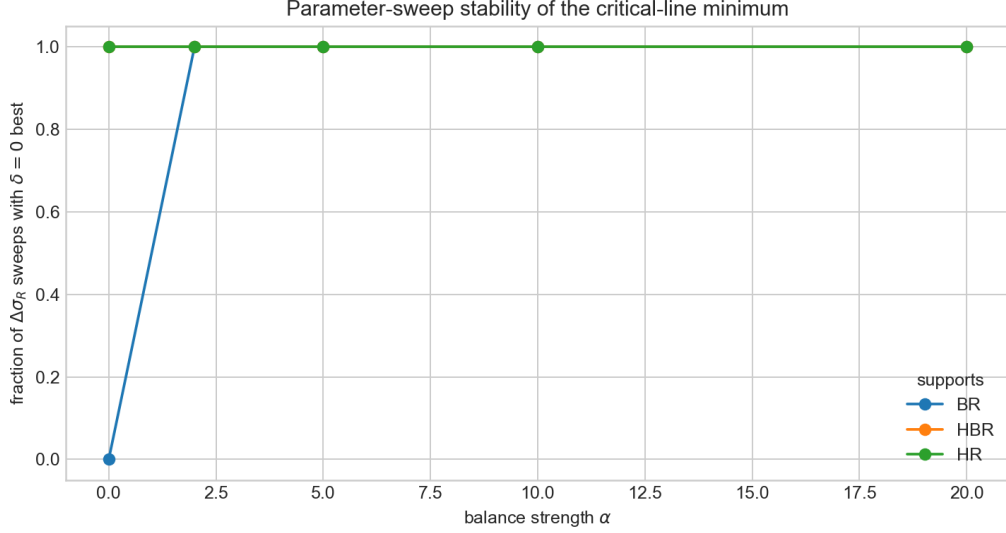


Figure 3: Parameter-sweep stability of the $\delta = 0$ minimum for known zeros. The full HBR and no-balance HR diagnostics are stable across the tested parameter grid.

Across the tested sweep, the full HBR and no-balance HR diagnostics select $\delta = 0$ for all tested $(\alpha, \Delta\sigma)$ combinations. The no-H BR diagnostic is less stable, selecting $\delta = 0$ in 80% of the tested cases.

4.4 Score Decomposition

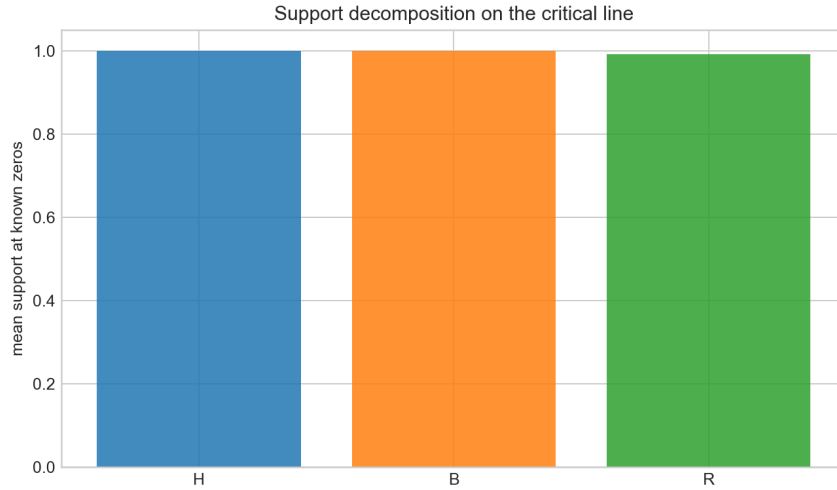


Figure 4: Mean diagnostic-term values at known zeros on the critical line. The H and B terms are essentially maximal by construction; the nontrivial variation is mainly in the local response term.

At known zeros, the H term is essentially maximal because $|\zeta(1/2 + it_n)| \simeq 0$. The B term is also maximal because $\sigma = 1/2$. This again shows why the result must be interpreted as a diagnostic score, not as a theorem.

5 Pair-Correlation Diagnostic

The previous sections test individual known zero ordinates. This is inherently limited because the H term directly rewards $|\zeta(s)| \simeq 0$. A more interesting diagnostic is therefore collective: does the geometry of the zero sequence itself resemble a known spectral reference distribution?

Let

$$\Delta t_n = t_{n+1} - t_n. \quad (10)$$

The spacings are unfolded using the leading local zero density,

$$s_n = \Delta t_n \frac{\log(t_n/2\pi)}{2\pi}, \quad (11)$$

and rescaled to unit mean. The empirical spacing distribution is compared with a unit-mean GUE nearest-neighbour spacing proxy using the Wigner-surmise form

$$P_{\text{GUE}}(s) = \frac{32}{\pi^2} s^2 \exp\left(-\frac{4s^2}{\pi}\right). \quad (12)$$

This is not the full pair-correlation theorem; it is a finite-sample nearest-neighbour diagnostic.

For histogram bins k , define

$$V_k^{\text{pair}} = \frac{1}{1 + |P_{\text{emp}}(s_k) - P_{\text{GUE}}(s_k)|}, \quad (13)$$

and a weighted pair score

$$V_{\text{pair}} = \sum_k P_{\text{GUE}}(s_k) V_k^{\text{pair}}. \quad (14)$$

The corresponding pair-geometry cost is

$$F_{\text{pair}} = -\log(\epsilon + V_{\text{pair}}). \quad (15)$$

Table 3: Pair-correlation diagnostic using the first 200 zeta zeros.

Series	Spacings	L^1 to GUE	V_{pair}	F_{pair}
Jittered zeta spacings	199	0.2059	0.9844	0.0157
Zeta normalised spacings	199	0.2938	0.9797	0.0205
Uniform-points control	199	0.8103	0.9573	0.0437
Poisson control	199	0.8225	0.9559	0.0451

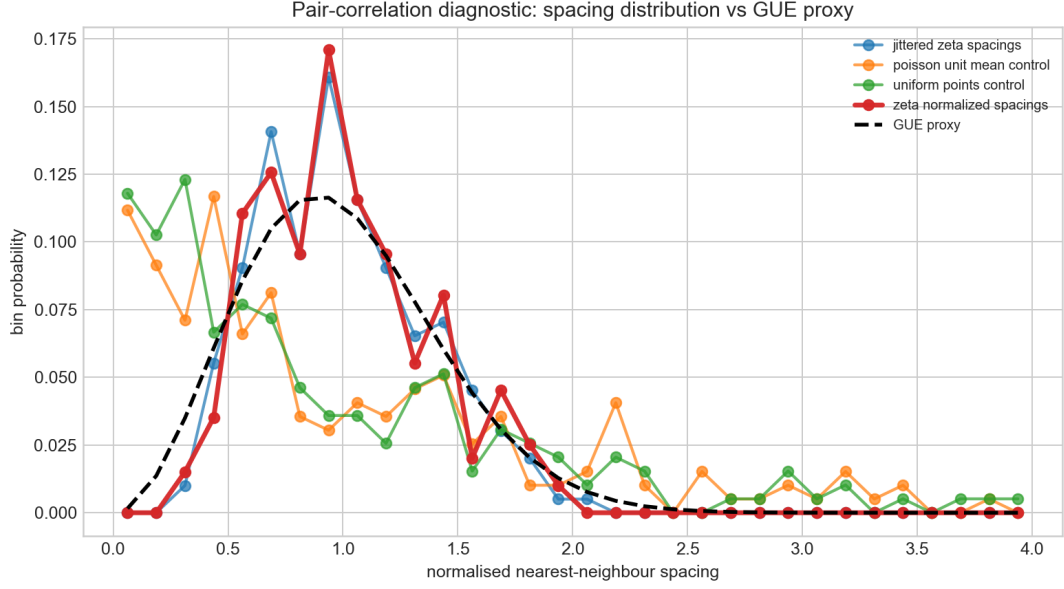


Figure 5: Normalised nearest-neighbour spacing distribution compared with the GUE spacing proxy. The zeta spacing geometry is visibly closer to the GUE proxy than the Poisson and uniform controls.

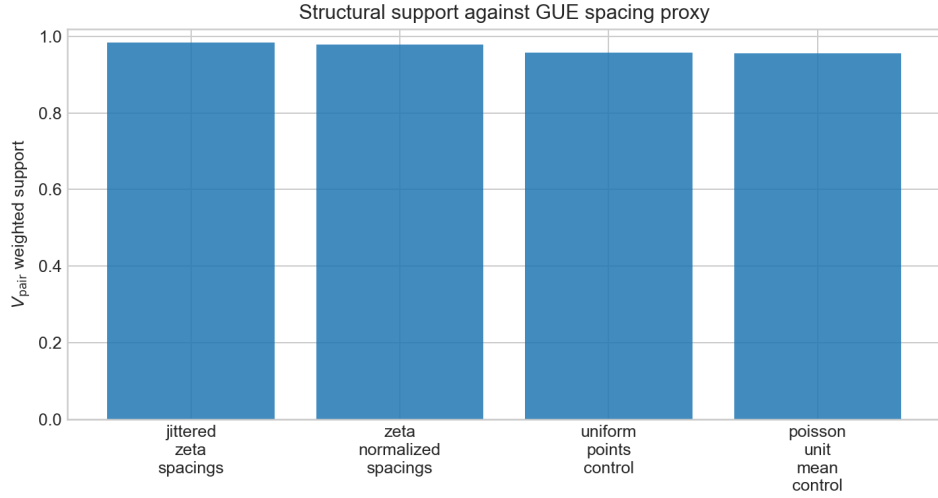


Figure 6: Pair-correlation diagnostic against the GUE spacing proxy. The zeta sequence and its small jittered variant have lower diagnostic stress, while Poisson and uniform controls are higher.

This pair-correlation test is less tautological than the individual-zero scan because it does not ask whether a point is a zero. It asks whether the spacing geometry of the zero sequence resembles the expected random-matrix structure. The pair diagnostic is conceptually closer to spectral-universality tests than to direct zero-location tests. The result is positive but still conservative: zeta spacings have lower diagnostic stress than Poisson and uniform controls, while the jittered-zero sequence remains slightly stronger in this finite histogram test. Thus the diagnostic detects a robust spacing-geometry mode rather than an exact zero-location theorem.

6 Interpretation

The legitimate conclusion is narrow:

Known zeta-zero locations exhibit a low-cost minimum on the critical line under the proposed diagnostic score.

The equally important limitation is:

The diagnostic partly builds in critical-line preference through the balance term and directly rewards known zeros through the zeta-null term.

Therefore this experiment does not provide evidence for the Riemann Hypothesis. It is better understood as a stress test of a heuristic scoring rule on a rigid mathematical object. Its value is methodological: it separates three things that are easy to confuse in exploratory work: score-induced structure, object-induced structure, and residual ambiguity after controls.

7 Limitations

- The H term directly encodes small $|\zeta|$.
- The B term explicitly favours $\sigma = 1/2$.
- The robustness term is heuristic and not derived from analytic number theory.
- Only a finite pointwise set of the first 40 zeros is tested.
- The pair-correlation component uses the first 200 zeros and should be viewed as a finite-sample diagnostic, not as an asymptotic statement.
- No statement is made about zeros beyond the tested range.
- No theorem, proof strategy, or implication for the Riemann Hypothesis is claimed.
- The pair-correlation diagnostic uses a Wigner-surmise nearest-neighbour proxy, not the full GUE spacing law or Montgomery pair-correlation formula.
- The jittered-zero control remains very close to the GUE proxy, showing that the pair diagnostic probes robust spacing geometry rather than exact zero locations.

8 Reproducibility

Repository folder:

experiments/zeta_critical_line_selection/

Main script:

```
python3 experiments/zeta_critical_line_selection/  
zeta_critical_line_selection_v02.py
```

The script writes:

- zeta ordinate tables,
- full score grids,
- delta summaries,

- parameter-sweep summaries,
- pair-correlation spacing summaries,
- figures,
- a JSON summary file.

9 Data and License Note

No external dataset file is redistributed. The script computes zeta zeros using `mpmath.zetazero` and evaluates the zeta function directly. Generated CSV, JSON, and PNG files are derived reproducibility artifacts.

10 Conclusion

This extra paper tests a simple diagnostic score on the critical line of the Riemann zeta function. Known zeros have low diagnostic stress at $\sigma = 1/2$, but the controls and ablations show that part of this preference is produced by the score design. The pair-correlation test adds a less pointwise check: zeta nearest-neighbour spacings have lower diagnostic stress against a GUE proxy than Poisson or uniform controls, but the signal remains robust under small jitter and should therefore be read as a spacing-geometry signal rather than exact zero-location evidence.

The result is not a Riemann-Hypothesis result. Its value is methodological: a score can look compelling, but only controls reveal how much of the structure was built in from the start. Even on one of the most rigid objects in mathematics, structural diagnostics require careful controls; otherwise they may tell us more about the diagnostic than about the object.

References

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- [3] A. M. Odlyzko, “On the distribution of spacings between zeros of the zeta function,” *Mathematics of Computation* **48**, 273–308 (1987).
- [4] F. Johansson et al., *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, <https://mpmath.org/>.